Vipul Sapra

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EDUCATION

University of Cincinnati, Carl H. Lindner College of Business — **Cincinnati, OH**
Master of Science in Information Systems and Graduate Certificate in Data Science — **April 2022**
Cumulative GPA: 3.95 / 4.0

Pune University, Maharashtra Institute of Technology — **Pune, India**
Bachelor of Engineering in Petrochemical — **June 2017**

TECHNICAL SKILLS

Concepts: Solution Architecture, Program Leadership, Predictive Modeling, Agentic AI & Prompt Engineering, Dimensional Data Modeling, Star/Snowflake Schemas, SCD Type 1/2, Data Governance & Quality Audits, ETL/ELT Pipeline Optimization, Root Cause Analysis, Capacity Planning, Workforce Optimization

Programming Languages & Tools: Azure (Data Factory, Synapse, SQL), AWS (S3, Redshift), Snowflake, Apache Spark, PySpark, Python, Scikit-Learn, XGBoost, SARIMA, Power BI, Tableau, MicroStrategy, Airflow, One Automation, KIRO IDE, SQL Stored Procedures

Certifications: Data Science Graduate Certificate – University of Cincinnati

PROFESSIONAL EXPERIENCE

Charter Communications — Charlotte, NC

Lead Solution Architect - AI, Data & Analytics Delivery — **May 2022 – Present**

- **Architected end-to-end enterprise data solutions** overseeing full lifecycle delivery of high-throughput data processing platforms, dimensional data modeling (Star/Snowflake schemas), and Slowly Changing Dimensions (SCD Type 1/2) ensuring 99.9% data availability SLA.
- **Led AI and predictive modeling initiatives** utilizing XGBoost, Random Forests, Scikit-Learn, and SARIMA forecasting to optimize workforce capacity planning and customer behavior modeling, successfully driving a 20% improvement in reliability and response times.
- **Spearheaded advanced Agentic AI frameworks** and automated QA testing systems such as SpectrumIQ and KIRO IDE prompt engineering, cutting manual operational turnaround by 50% and accelerating enterprise data delivery pipelines by 40%.
- **Engineered automated data governance frameworks** including proactive anomaly detection machine learning systems (95% accuracy) and Automated Data Quality Audit firewalls that reduced operational discrepancies and turnaround times by over 80%.
- **Directed cross-functional technical teams and client management** by establishing production runbooks, Airflow batch monitoring, and Root Cause Analysis postmortems, aligning executive KOM dashboards and cloud deployments with strategic business goals.

Master of Science in Information Systems & Data Science Graduate — **May 2022 – Present**

- **Completed advanced academic training** at the University of Cincinnati with a 3.95/4.0 GPA, specializing in Information Systems and earning a Graduate Certificate in Data Science.
- **Mastered complex quantitative methodologies** spanning database architectures, predictive analytics, and enterprise data governance frameworks.
- **Applied cutting-edge data science theories** to solve simulated enterprise data challenges and build scalable machine learning prototypes.

Bachelor of Engineering in Petrochemical — **May 2022 – Present**

- **Built rigorous analytical foundation** at Pune University MIT through complex process engineering, mathematical modeling, and systematic problem-solving.
- **Developed strong cross-functional competencies** in optimizing operational workflows and managing technical project deliverables.
- **Cultivated core engineering principles** applied later to large-scale data systems and modern cloud data architectures.

Ameritas Insurance — Cincinnati, OH

Data Governance Intern — **September 2021 – April 2022**

- **Co-engineered the enterprise-wide Data Advocacy & Governance Program** across insurance portfolios, establishing foundational frameworks for data democratization and organizational accountability.
- **Architected a comprehensive Data Intelligence ecosystem in Collibra**, establishing a centralized Data Dictionary, automated Data Quality rules, and end-to-end **Data Lineage** to ensure regulatory compliance and transparency for financial and insurance assets.

Accenture — Pune, India

Test Engineering Analyst — **September 2017 – December 2020**

- **Orchestrated end-to-end Quality Assurance and system integration testing** for mission-critical client applications at **Allianz Insurance**, leveraging Agile SDLC methodologies to ensure rapid, zero-defect feature deployments.
- **Spearheaded the Health Insurance module QA strategy**, directing cross-functional efforts for regression testing, test data validation, and the seamless rollout of complex new business functionalities.
- **Drove operational excellence by automating core regression test suites**, significantly increasing engineering productivity and reducing manual testing overhead through scalable automation frameworks.